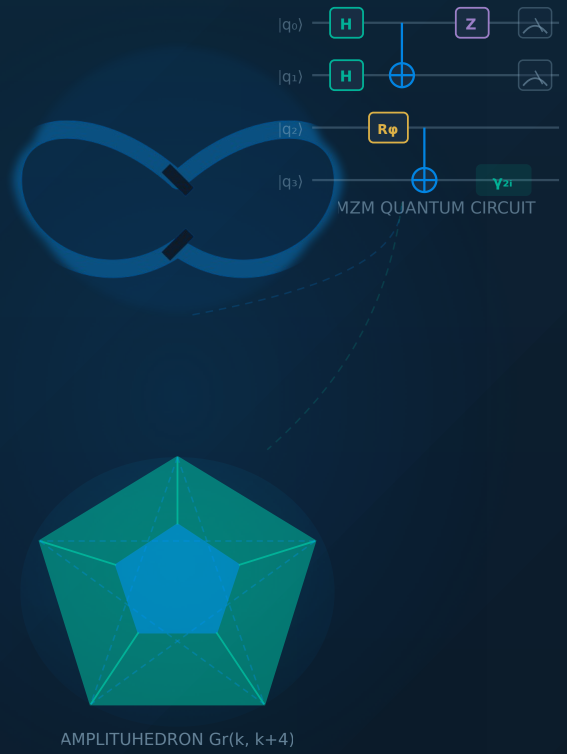
Faddeev – Amplituhedron –MZM

From Knot Topology to Quantum Computation
via Dilogarithm Geometry

Pentagon Identity → **Categorification** → **8-Manifold** → **Qubit Encoding**

☆ **4 Stages · 6 Months · Novel Mathematics → Quantum Advantage**

RQ
Robert Quinn, MBA
Principal Investigator · Program Director



Pentagon Identity

$\Phi_b(p)\, \Phi_b(p+q)\, \Phi_b(q) = \Phi_b(q)\, \Phi_b(p)$

KNOT INVARIANTS · 3-MANIFOLDS · QUANTUM

CORE HYPOTHESIS

“The **Pentagon Identity** of the Faddeev Dilogarithm is the **universal organizing principle** connecting hyperbolic 3-manifold invariants, higher categorical structures, amplituhedron geometry, and **topologically protected quantum computation.**”

Research Program Thesis · Robert Quinn, PI

Knot Complement ↔ State Integral

$Z_M(\tau)$ encodes hyperbolic volume via Φ_b tetrahedra product

Pentagon ↔ Braiding Coherence

$\Phi_b(p)\Phi_b(q) = \Phi_b(q)\Phi_b(p+q)\Phi_b(p)$ governs A_∞ categorical morphisms

Amplituhedron ↔ Quantum Optimizer

$Gr_{\mathbb{H}}(k, k+4)$ geometry drives QAOA & GQPSO landscape

RESEARCH ARCHITECTURE

Four Stages of Research

Months 1–6 · Sequential Progression

STAGE 01 — MONTHS 1-2

Foundational

Faddeev Dilogarithm

Extend $\Phi_b(x)$ to matrix & quaternion (H) modules. Prove noncommutative pentagon identity. Validate state integrals $Z_M(\tau)$ on hyperbolic knot complements $\mathbf{4_1}$ and $\mathbf{5_2}$ via SnapPy triangulations.

q-series to order q^{20} · SymPy + mpmath

Quantum Topology Foundation

STAGE 02 — MONTHS 2-4

Algebraic Lift

Categorification

Categorify the pentagon identity as natural transformations in A_∞-categories. Deform Neumann-Zagier matrices via $\mathbf{A_\theta = A + \theta B}$ with $\theta \in$ H. Compute DT invariants as cochain complexes → 4D topological indices.

Twisted NZ matrices · Loop/membrane excitations

Higher Categorical Algebra

STAGE 03 — MONTHS 4-5

Geometric Model

8-Manifold Amplituhedron

Extend Gr(k, k+4) over $\mathbb{R}$ → $\mathbf{Gr_{\mathbb{H}}(k, k+4)}$ over quaternions, yielding an 8-manifold. BCFW tetrahedral triangulations (Even-Zohar 2025). RJMCMC Monte Carlo sampling. Apply to Kagome/moiré BEC vortex lattices; recover BKT transition.

4×A100 GPU · RJMCMC chains · BKT topology

Geometric Realization

STAGE 04 — MONTH 5-6

Algorithmic

Quantum Computation

GQPSO-CCEA & QAOA with NZ locality constraints; beat Goemans-Williamson 0.878. QSVT block-encode DT Hamiltonian (error $<10^{-4}$). Kitaev chain **MZM qubit** prototypes targeting fidelity ≥ 0.99 on IBM 127-qubit Eagle QPU.

IBM Eagle 127q · QSVT · MZM fidelity ≥ 0.99

Quantum Implementation

STAGE 1 OF 4 · FOUNDATIONAL

Formalizing $\Phi_b(x)$ on Quaternion Modules

⬡ **PENTAGON IDENTITY**

$\Phi_b(p)\,\Phi_b(q) = \Phi_b(q)\,\Phi_b(p+q)\,\Phi_b(p)$

BOX 1 — MATHEMATICAL GOAL

Quaternionic Extension of $\Phi_b(x)$

Extend **$\Phi_b(x)$** to **ℍ-valued operators** on quaternion modules. Prove the functional equation in the **noncommutative** setting:

$$\Phi_b(p)\,\Phi_b(-p) = e_{\pi i p^2} \cdot \Phi_b(0)^2$$

Noncommutative · ℍ-modules

BOX 2 — COMPUTATIONAL TARGET

Numerical Integration Stack

[redacted] noncommutative algebra engine for symbolic manipulation. [redacted] arbitrary-precision numerical integration of Φ_b. **SnapPy** triangulation data pipeline for the **4_1 figure-eight knot** — NZ matrix extraction and gluing equations.

SymPy · mpmath · SnapPy

BOX 3 — KEY DELIVERABLE

q-Series Coefficients to Order q^{20}

Reproduce **q-hypergeometric series** coefficients for knots **4_1** and **5_2** matching Garoufalidis–Kashaev (2013, arXiv:1304.2705). Verification target: exact agreement to order [redacted] for both knot complements. Forms the quantitative benchmark for Stage 1 success.

4_1 knot · 5_2 knot · GK 2013

BOX 4 — PHYSICAL TARGET

State Integral on Hyperbolic Knot Complements

Validate the **state integral model** on hyperbolic 3-manifolds:

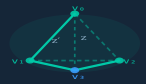
$$Z_M(\tau) = \int \prod \Phi_b(x_i)\, dx_i$$

Confirm topological invariance under change of triangulation for 4_1 and 5_2 complements.

4_1 FIGURE-EIGHT KNOT

Hyperbolic knot · 4 crossings
Vol ≈ 2.0298 (complete)

IDEAL TETRAHEDRON — NZ MATRIX



Ideal tetrahedra · shape params z, z', z''
gluing → NZ matrix [A|B]

KEY REFERENCE
Garoufalidis–Kashaev
arXiv:1304.2705 (2013)
Target: q^{20} agreement

STAGE 2 · MONTHS 4–5

Stage 2: **Lifting to Higher Algebra**

Classical Framework | Categorified Extension | Lift / Upgrade

CLASSICAL
Scalar / Numerical Invariants

CATEGORIFIED
A∞-Structures · Quaternionic · Complexes

PENTAGON IDENTITY

Scalar Functional Equation

The pentagon relation holds as a numerical identity between complex-valued functions of a single variable

$\Phi\_b(p)\cdot\Phi\_b(q) = \Phi\_b(r)\cdot\Phi\_b(s)\cdot\Phi\_b(t) \in \mathbb{C}$

LIFT →

PENTAGON AS 2-MORPHISM

Natural Transformation of Functors

Pentagon becomes a coherence condition — a natural transformation between composition functors in an A∞-category

$\mu^2 \circ (\mu^2 \otimes \mathrm{id}) \Rightarrow \mu^2 \circ (\mathrm{id} \otimes \mu^2)$ in A∞-Cat

NEUMANN-ZAGIER MATRICES

Integer Gluing Data

Matrices $A, B \in \mathrm{Mat}(n\times n, \mathbb{Z})$ encode hyperbolic ideal triangulation gluing equations for knot complements

$A\cdot z + B\cdot\bar{z} = 0$ (gluing eqs)

TWIST →

TWISTED NZ MATRICES

Quaternionic Deformation A_θ

Deform integer matrices by quaternionic parameter $\theta \in \mathbb{H}$, encoding 4D non-commutativity into gluing data

$A\_\theta = A + \theta B, \quad \theta \in \mathbb{H}$ (noncommutative)

DONALDSON-THOMAS INVARIANT

A Single Number

DT invariant counts ideal sheaves; output is an integer or rational number

$DT(X) = \chi(\mathrm{Hilb}_n X) \in \mathbb{Z}$

CAT. →

CATEGORIFIED DT INVARIANT

Cochain Complex — Euler Char. Recovers Classical

DT is replaced by a cochain complex $C^\bullet$ with differential; the integer is recovered as its Euler characteristic

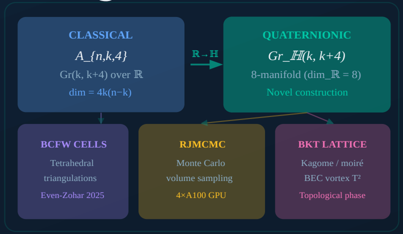
$\chi(C^\bullet) = \Sigma(-1)^i \dim H^i(C^\bullet) = DT(X)$

4D TARGET | Map categorified pentagon to **4D topological indices** — pentagon & hexagon equations governing **loop / membrane excitations** in exotic topological orders (Huang–Zhang–Ye, JHEP 2025)

STAGE 3 OF 4

Quaternionic Amplituhedron from Tetrahedral Triangulations

Extending Arkani-Hamed & Trnka to 8 real dimensions



4 × A100 GPU Parallel MCMC chains

KEY TECHNICAL PILLARS

Classical Amplituhedron Baseline

$\mathcal{A}_{n,k,4}$ lives in **Gr(k, k+4) over** ℝ — encodes N=4 SYM scattering amplitudes as a geometric volume form, bypassing Feynman diagrams entirely.

Our Extension: ℝ → ℍ Quaternionic Grassmannian

Replace scalars with **quaternions H** → $Gr_{\mathbb{H}}(k, k+4)$, naturally yielding an **8-manifold** ($\dim_{\mathbb{R}} \mathbb{H} = 4$), doubling the geometric richness of the classical construction.

BCFW Cells from Tetrahedral Triangulations

Even-Zohar et al. (2025, **Inventiones**) proved the BCFW tiling conjecture. We adapt tetrahedral triangulation cells to tile $Gr_{\mathbb{H}}$, giving a rigorous combinatorial decomposition.

RJMCMC Discretization for Volume Sampling

Reversible-Jump MCMC enables trans-dimensional Monte Carlo sampling of the **amplituhedron volume form** across BCFW cells — parallelized across **4×A100 GPUs** for scalable throughput.

Application: Kagome/Moiré BEC Vortex Lattices → BKT Transition

Triangulate magnetic unit cell as **T^2**; sample vortex configurations via RJMCMC; recover the **Berezinskii-Kosterlitz-Thouless** topological phase transition in 2D superfluids from amplituhedron geometry.

Stage 4: Hybrid Classical-Quantum Optimization

GQPSO-CCEA · QAOA · QSVT · MZM

BOX 1

QAOA

- Hamiltonian H_C from NZ matrix elements J_{ij} with locality constraints from NZ connectivity graph
- p-layer depth analysis $p = 1 \ldots 8$ targeting performance beyond Goemans-Williamson bound
- Beating GW 0.878 approximation threshold on NZ-structured MAX-CUT instances

Target: GW > 0.878 bound

BOX 2

GQPSO-CCEA

- Quantum-behaved PSO using Faddeev Φ_b as the quantum potential well guiding particle updates
- Coevolutionary subpopulations (CCEA) partitioned across Stage 3 amplituhedron landscape regions
- Parallel MCMC chains coupled with swarm attractors for quaternionic Grassmannian volume sampling

Potential well: $\Phi_b(x)$ landscape

BOX 3

QSVT Block-Encoding

- Quantum Singular Value Transformation (Martyn et al. 2021) unifying amplitude estimation, phase estimation & linear systems
- Block-encode Stage 2 DT Hamiltonian into unitary oracle via qubitized quantum walks
- Singular value approximation error target $< 10^{-4}$ with optimized polynomial degree

Error $< 10^{-4}$ on DT spectrum

BOX 4

MZM Qubits

- Kitaev chain model: logical qubit encoded as degenerate pair $\gamma_{2i-1}, \gamma_{2i}$ of Majorana zero modes
- Non-local encoding provides intrinsic topological protection against local perturbations & dephasing
- Target gate fidelity ≥ 0.99 under depolarizing noise model at $p = 10^{-3}$

Fidelity ≥ 0.99 · noise $p = 10^{-3}$

COMPUTATIONAL PLANNING

Resource **Requirements**

TOTAL BUDGET
**$85,00
0**
est. cloud compute

~7,700	CPU-hours	TOTAL CLASSICAL COMPUTE	
~1,200	GPU-hours (A100)	GPU ACCELERATION (STAGE 3–4)	
~500	QPU-hours	QUANTUM PROCESSING UNITS	
127	qubits	TARGET QPU SCALE (IBM EAGLE)	
$85,000	estimated	CLOUD COMPUTE BUDGET	
24	weeks	TOTAL SPRINT TIMELINE	

PER-STAGE COMPUTE BREAKDOWN

PU-hrs PU-hrs PU-hrs

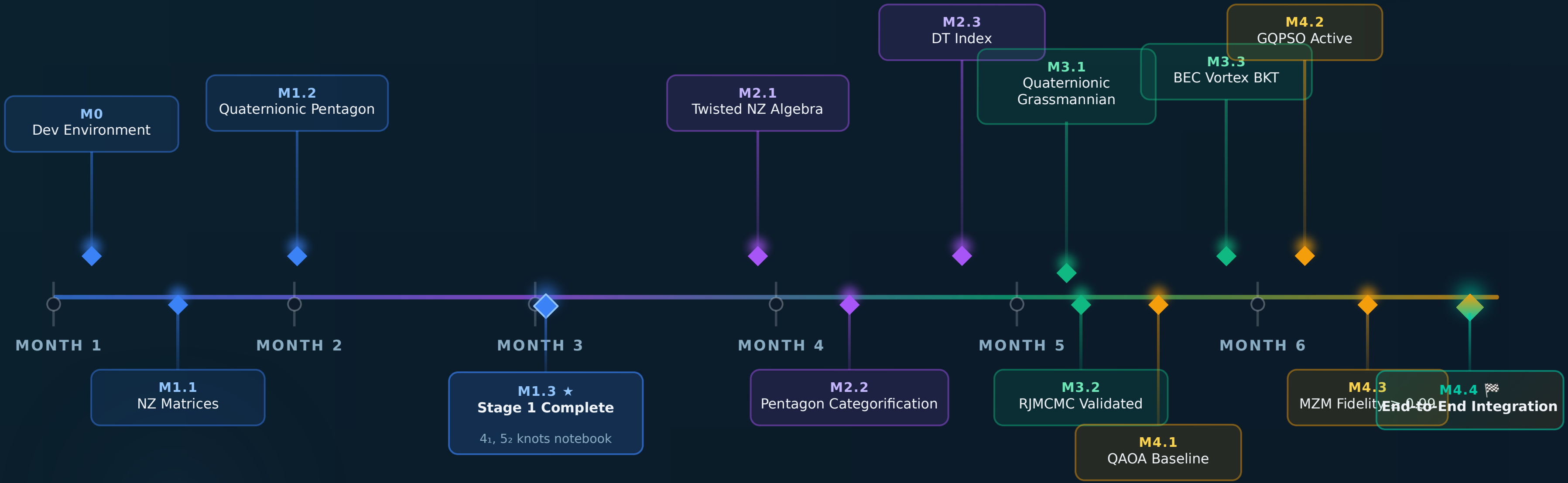
STAGE	CPU-HOURS	CPU SHARE	GPU-HOURS	GPU SHARE	QPU-HOURS	QPU SHARE
Stage 1 — Foundational Faddeev dilogarithm · Knot complements	**200**	3%	—		—	
Stage 2 — Algebraic Lift Categorification · Twisted NZ matrices · A∞ algebras	**500**	6%	—		—	
Stage 3 — Geometric Model 8-manifold amplituhedron · RJMCMC · BEC vortex lattices	**2,000**	26%	**800**	67%	—	
Stage 4 — Algorithmic GQPSO-CCEA · QAOA · QSVT · MZM qubit prototypes	**5,000**	65%	**400**	33%	**500**	100%
✓ **Totals**	**7,700**	100%	**1,200**	100%	**500**	100%

AWS EC2 c6a.48xlarge @ $7.22/hr · p4d.24xlarge @ $32.77/hr · IBM Quantum Premium @ ~$1.60/QPU-sec

Six-Month Milestone Roadmap

Sequential research stages · 24 sprints · Integrated delivery

STAGE 1 STAGE 2 STAGE 3 STAGE 4



STAGE 1 · MONTHS 1-3
Faddeev Dilogarithm Foundations

STAGE 2 · MONTHS 4-5
Categorification & Twisted NZ

STAGE 3 · MONTHS 5-6
8-Manifold Amplituhedron

STAGE 4 · MONTHS 5-6
Quantum Algorithms & MZM

12
MILESTONES

NOVELTY & SIGNIFICANCE

What We Do **vs.** State of the Art

ORIGINAL CONTRIBUTIONS

STATE OF THE ART *Current Limitations*		OUR CONTRIBUTION *Novel Advances*
✕ Dilogarithm over $\mathbb{C}$ only — $\Phi b(x)$ restricted to complex scalars; no noncommutative extension	VS	✓ Dilogarithm extended to $\mathbb{H}$-operators — $\Phi b(x)$ on quaternionic modules; noncommutative functional equation proved
✕ Pentagon as scalar equation — functional identity treated as a numerical relation between complex functions		✓ Pentagon as 2-categorical natural transformation — lifts to $A\infty$ functor coherence; encodes braiding in higher category
✕ Amplituhedron over real Gr(k, k+4) — scattering geometry confined to 4-real-dimensional real Grassmannian		✓ 8-manifold amplituhedron over Gr$\mathbb{H}$(k, k+4) — quaternionic extension doubles dimension; BCFW-triangulated via RJMCMC
✕ QAOA with generic Hamiltonians — circuit design decoupled from underlying topological structure		✓ QAOA constrained by topological NZ matrix structure — Neumann-Zagier locality encodes hyperbolic geometry into circuit depth
✕ MZM qubits in isolation — Majorana zero modes studied independently of knot/manifold invariants		✓ Full pipeline: knot invariant → topological qubit — end-to-end integration across all four mathematical domains

☆ **First unified framework connecting hyperbolic 3-manifold arithmetic to topologically protected quantum computation — across four independent domains simultaneously.**

PERSONNEL & NETWORK

Research Team & Collaborator Network

PI: Robert Quinn, MBA

CORE RESEARCH TEAM

RQ **Robert Quinn** PI
Mathematical Physics
Program design · Grant management · Mathematical physics
Stage 1 | Stage 4 | Lead

PA **Postdoc A** POSTDOC
Numerical Methods · HPC
Faddeev integrals · MCMC sampling · HPC pipelines
Stage 1 | Stage 3

PB **Postdoc B** POSTDOC
Algebraic Geometry
Tensor networks · QAOA circuits · Categorification
Stage 2 | Stage 4

GS **Grad Student** PHD
Coding · Testing · Docs
Notebook documentation · Validation · Integration testing
Stage 1 | Stage 3 | Stage 4

EXTERNAL COLLABORATOR NETWORK

5 World-Class Experts

GT **Stavros Garoufalidis** Georgia Tech — Quantum Topology
Quantum dilogarithm · State integrals · q-series knot invariants

AU **Jørgen Andersen** Aarhus University — Kashaev Invariants
Andersen-Kashaev TQFT · Quantum hyperbolic geometry · Volume conjecture

HV **Lauren Williams** Harvard University — Positroid / Amplituhedron
Positroid cells · Total positivity · Grassmannian combinatorics

USC **Peng Ye** Univ. of Southern California — 4D Topological Orders
Pentagon equations · Loop/membrane excitations · JHEP 2025 (Huang-Zhang-Ye)

MS **Chris Nayak** Microsoft Station Q — MZM / Topological Qubits
Non-Abelian anyons · Majorana zero modes · Topological quantum computation

Cross-disciplinary network spanning quantum topology · categorical algebra · topological computation · amplituhedron geometry

4 Core Members | 5 External Collaborators | 5 Institutions | 4 Research Stages

GRANT PROPOSAL

Funding Request & Broader Impact

$1.2M Total

3-Year NSF / DOE Research Program
PI: Robert Quinn, MBA · Lacey, NJ

BUDGET ALLOCATION

Personnel	40%
HPC / QPU Cloud	25%
Travel / Collab	20%
Indirect	15%

NSF DMS/PHY · DOE ASCR

THREE PILLARS OF BROADER IMPACT

FOUNDATIONAL MATHEMATICS

Quaternionic Pentagon Identity & Matrix State Integrals

First proof of quaternionic pentagon identity and matrix-valued state integrals — direct impact on quantum topology and low-dimensional geometry.

QUANTUM COMPUTING

Novel MZM Encoding from Topological Invariants

Path to fault-tolerant qubits with mathematical error-correction guarantees — Majorana zero mode protocol derived from knot complement arithmetic.

CONDENSED MATTER

RJMCMC-Amplituhedron for Kagome/Moiré BEC Vortex Lattices

First RJMCMC-amplituhedron simulation framework for topological superconductor design — new computational tool for BKT transition modeling.

"Mathematics that computes. Topology that protects."

Robert Quinn, MBA Principal Investigator
Submitted May 2026